THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 13

Delta Lake vs Parquet vs Iceberg — A Layered Comparison

SECTION 01

They're Not Really Competitors, They're Layers

A common interview mistake is treating Parquet, Delta Lake, and Iceberg as three interchangeable options at the same level. In reality, they build on top of each other. Both Delta Lake and Iceberg store their actual data as Parquet files; the difference is entirely in the metadata layer added on top.

CORE CONCEPT

Apache Iceberg and Delta Lake are both open lakehouse table formats that store data in Parquet on object storage with ACID transactions. The real difference between them is metadata strategy and ecosystem, not the underlying file format.

SECTION 02

Layer 1: Apache Parquet

Parquet is a columnar file format, not a table format. It is efficient for storage and analytical reads, but on its own it has no transaction log, no ACID guarantees, and no built-in time travel.

- No concurrency control: simultaneous writes can corrupt data.
- No schema safety net: schema evolution is limited, and changes often require rewriting data.
- No table-level metadata: the raw files themselves are the only source of truth.

SECTION 03

Layer 2: Delta Lake

Delta Lake wraps Parquet files with a transaction log, delivering everything covered across this series' Delta Lake phase: ACID transactions, schema enforcement, time travel, and efficient data layout via OPTIMIZE and Liquid Clustering.

HOW IT TRACKS CHANGES

Delta Lake uses a flat transaction log stored in a _delta_log directory. Each transaction creates a JSON file, and periodic Parquet checkpoints consolidate the log for faster reads. It is deeply integrated with Apache Spark, where it delivers its strongest performance and feature support.

SECTION 04

Layer 3 (Alternative): Apache Iceberg

Iceberg solves a similar problem to Delta Lake, ACID guarantees and reliability on top of Parquet, but using a different metadata design, and with a different founding philosophy.

HOW IT TRACKS CHANGES

Iceberg uses a multi-layered metadata tree: a catalog points to a metadata file, which references a manifest list, which references individual manifests that track the actual Parquet data files. Each manifest carries column-level statistics, enabling aggressive file pruning at query planning time, and Iceberg never needs to list files in object storage, since all file tracking is done through this metadata tree.

- Multi-engine by design: Spark, Trino, Flink, Snowflake, and Athena can all read and write the same Iceberg table.
- Partition evolution: changing a partitioning strategy is a metadata-only operation in Iceberg, whereas Delta Lake has historically required a fuller data rewrite for the same change.

SECTION 05

Side-by-Side Comparison

Aspect	Parquet	Delta Lake	Iceberg
What It Is	A file format	A transactional layer on Parquet	A transactional layer on Parquet
Metadata Structure	None (files only)	Flat JSON log + Parquet checkpoints	Multi-layered metadata tree (Avro)
ACID Transactions	No	Yes	Yes
Time Travel	No	Yes	Yes
Primary Ecosystem	Engine-agnostic	Spark / Databricks	Multi-engine, vendor-neutral
Partition Changes	Requires rewrite	Often requires rewrite	Metadata-only operation

SECTION 06

The 2026 Trend: Convergence

Rather than one format simply winning, 2026 has seen active convergence between Delta Lake and Iceberg, driven largely by customer demand for interoperability across tools.

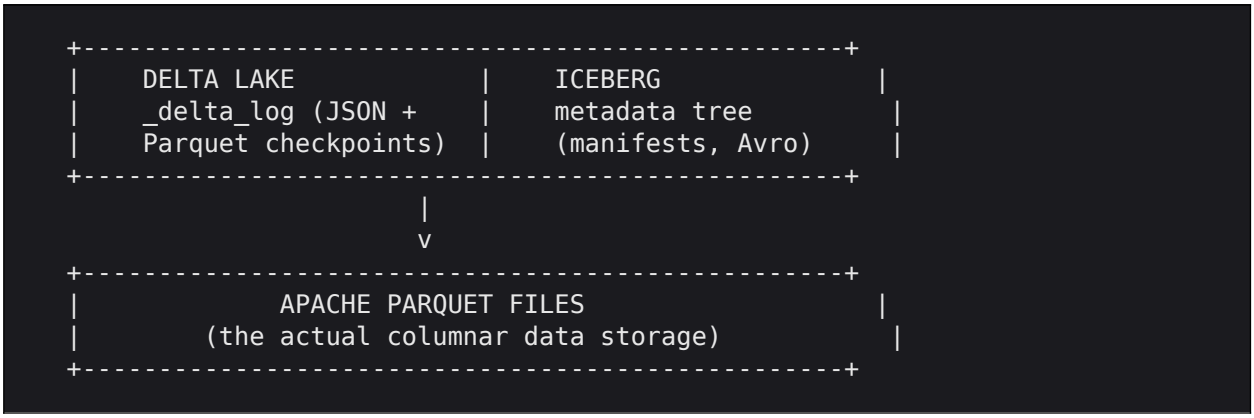
- Delta UniForm lets Delta tables be read as Iceberg tables by other engines like Snowflake, reducing the need to maintain duplicate copies of the same data in two formats.
- Iceberg has adopted several Delta-originated features in its newer specification versions, narrowing the technical gap between the two.

PRACTICAL TAKEAWAY

Choosing a table format is increasingly a workload and ecosystem decision, not a permanent architectural commitment. Teams standardized on Databricks and Spark generally get the most out of Delta Lake, while teams needing true multi-engine flexibility often lean toward Iceberg.

SECTION 07

The Layered Stack



SECTION 08

Glossary — Day 13 Terms

Term	Definition
Apache Parquet	A columnar file format optimized for analytical reads; the underlying storage format for both Delta Lake and Iceberg.
Table Format	A metadata layer added on top of raw data files to provide ACID transactions, schema management, and time travel.
Manifest File	In Iceberg, a file tracking a subset of data files along with column-level statistics, used for query pruning.
Partition Evolution	The ability to change a table's partitioning strategy without rewriting existing data, native to Iceberg's metadata design.
Delta UniForm	A Databricks feature that lets Delta tables also be read as Iceberg tables by other engines.

SECTION 09

Interview Questions & Answers

Q1. What is the relationship between Parquet, Delta Lake, and Iceberg?

A. Parquet is the underlying file format. Delta Lake and Iceberg are both transactional metadata layers built on top of Parquet files, adding ACID transactions, schema management, and time travel; they are alternative approaches to the same underlying problem, not a strict hierarchy.

Q2. What is the core architectural difference between Delta Lake and Iceberg?

A. Delta Lake uses a flat transaction log of JSON files with periodic Parquet checkpoints. Iceberg uses a multi-layered metadata tree of manifest files and manifest lists in Avro format. Both achieve similar guarantees through different metadata designs.

Q3. Why might a team choose Iceberg over Delta Lake?

A. Primarily for multi-engine flexibility, since Iceberg was designed vendor-neutral from the start and is broadly supported across Spark, Trino, Flink, Snowflake, and Athena, avoiding lock-in to a single ecosystem.

Q4. Why might a team choose Delta Lake over Iceberg?

A. Because Delta Lake is deeply integrated with Apache Spark and the Databricks ecosystem, delivering its strongest performance and feature support there, which suits teams already standardized on that stack.

Q5. What is Delta UniForm?

A. A Databricks feature that allows Delta tables to also be read as Iceberg tables by other engines like Snowflake, addressing interoperability without requiring teams to maintain duplicate copies of data in two separate formats.

Q6. Does Parquet on its own provide ACID guarantees or time travel?

A. No. Parquet is only a file format; it has no transaction log, no built-in ACID guarantees, and no time travel. Those capabilities come from a table format like Delta Lake or Iceberg layered on top.

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